

organelles membrane-bound cell compartments

translation synthesis of protein using mRNA code

prokaryotes single-celled organisms without a nucleus

eukaryotes organisms with cells possessing a nucleus

mitochondria energy-producing cell organelle

chloroplasts the photosynthetic organelles of plants and algae

catalytic describes a substance that speeds a reaction without being consumed

Bibliography

Kazazian, Haig H., Jr. "L1 Retrotransposons Shape the Mammalian Genome." *Science* 289, no. 5482 (2000): 1152–1153.

Varmus, H. "Reverse Transcription." *Scientific American* 257, no. 3 (1987): 56–59.

Ribosome

Ribosomes are the cellular **organelles** that carry out protein synthesis, through a process called **translation**. They are found in both **prokaryotes** and **eukaryotes**, these molecular machines are responsible for accurately translating the linear genetic code, via the messenger RNA, into a linear sequence of amino acids to produce a protein. All cells contain ribosomes because growth requires the continued synthesis of new proteins. Ribosomes can exist in great numbers, ranging from thousands in a bacterial cell to hundreds of thousands in some human cells and hundreds of millions in a frog ovum. Ribosomes are also found in **mitochondria** and **chloroplasts**.

Structure

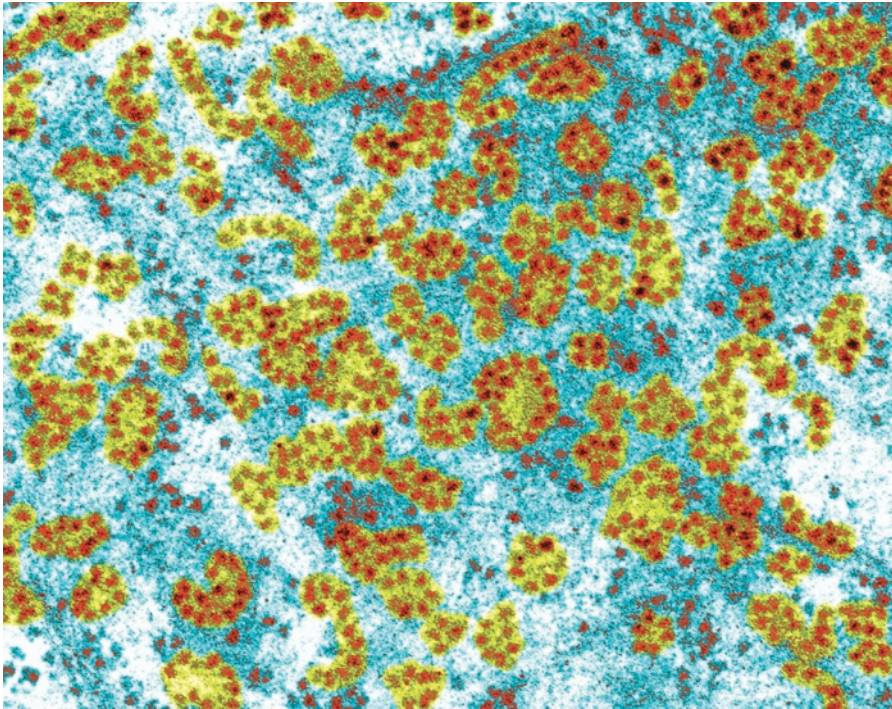
The ribosome is a large ribonucleoprotein (RNA-protein) complex, roughly 20 to 30 nanometers in diameter. It is formed from two unequally sized subunits, referred to as the small subunit and the large subunit. The two subunits of the ribosome must join together to become active in protein synthesis. However, they have distinguishable functions. The small subunit is involved in decoding the genetic information, while the large subunit has the catalytic activity responsible for peptide bond formation (that is, the joining of new amino acids to the growing protein chain).

In prokaryotes, the small subunit contains one RNA molecule and about twenty different proteins, while the large subunit contains two different RNAs and about thirty different proteins. Eukaryotic ribosomes are even more complex: the small subunit contains one RNA and over thirty proteins, while the large subunit is formed from three RNAs and about fifty proteins. Mitochondrial and chloroplast ribosomes are similar to prokaryotic ribosomes.

In spite of its complex composition, the architecture of the ribosome is very precise. Even more remarkable, ribosomes from all organisms, ranging from bacteria to humans, are very similar in their form and function. Recent breakthroughs in studies of ribosome structure, using techniques such as scanning, cryo-electron microscopy, and X-ray crystallography, have provided scientists with highly refined structures of this complex organelle. One particularly exciting conclusion from studies of the large subunit is that it is ribosomal RNA (rRNA), and not protein, that provides the **catalytic** activity for peptide bond formation. That is, it forms the chemical linkage between the amino acids of the growing protein molecule.

Synthesis

The synthesis of ribosomes is itself a very complex process, requiring the coordinated output from dozens of genes encoding ribosomal proteins and rRNAs. Ribosomes are assembled from their many component parts in an orderly pathway. In eukaryotes, rRNA synthesis and most of the assembly



Ribosomes from a liver cell are represented by the darkened areas of the magnified image. These organelles contain RNA and use it for protein synthesis.

steps occur in a structure within the nucleus called the nucleolus. Eukaryotic ribosome synthesis is especially complicated, because the ribosomal proteins themselves are made by ribosomes in the cytoplasm (that is, outside of the nucleus), so they then must be imported into the nucleolus for assembly onto the nucleolus-derived rRNA. Once assembled, the nearly complete ribosomal subunits are then exported out of the nucleus and back into the cytoplasm for the final steps of assembly.

The exact details of the **in vivo** ribosome assembly pathway (the process of ribosome assembly within the living cell) are still under investigation. Assembly in eukaryotic cells involves not only the components of the mature particles, but also dozens of auxiliary factors that promote the efficient and accurate construction of the ribosome during its assembly. However, bacterial ribosomes can be constructed **in vitro** using purified ribosomal proteins and rRNAs. These ribosomes appear to function normally in in vitro translation reactions.

Ribosome Function

Translation of messenger RNA (mRNA) by ribosomes occurs in the cytoplasm. In bacterial cells, ribosomes are scattered throughout the cytoplasm. In eukaryotic cells, they can be found both as free ribosomes and as bound ribosomes, their location depending on the function of the cell. Free ribosomes are found in the cytosol, which is the fluid portion of the cytoplasm, and are responsible for manufacturing proteins that will function as soluble proteins within the **cytoplasm** or form structural elements, including the cytoskeleton, that are found within the cytosol.

Bound ribosomes are attached to the outside of a membranous network called the **endoplasmic reticulum** to form what is termed the “rough” endoplasmic reticulum. Proteins made by bound ribosomes are intended to be

in vivo “in life”; in a living organism, rather than in a laboratory apparatus

in vitro “in glass”; in lab apparatus, rather than within a living organism

cytoplasm the material in a cell, excluding the nucleus

endoplasmic reticulum network of membranes within the cell

incorporated into membranes, or packaged for storage, or exported outside of the cell. Ribosomes exist either as a single ribosome (that is, one ribosome translating an mRNA) or as polysomes (two or more ribosomes sequentially translating the same mRNA in order to make multiple copies of the same protein).

Ribosomes have the critical role of mediating the transfer of genetic information from DNA to protein. Ribosomes translate this code using an intermediary, the messenger RNA, which is a copy of the DNA that can be interpreted by ribosomes. To begin translation, the small subunit first identifies, with the help of other protein factors, the precise point in the RNA sequence where it should begin linking amino acids, the building blocks of protein. The small subunit, once bound to the mRNA, is then joined by the large subunit and translation begins. The amino acid chain continues to grow until the ribosome reaches a signal that instructs it to stop.

Many of the antibiotics used in humans and other animals to treat bacterial infections specifically inhibit ribosome activity in the disease-causing bacteria, without affecting ribosome function in the host-animal's cells. These antibiotics work by binding to a protein or RNA target in the bacterial ribosome and inhibiting translation. In recent years, the misuse of antibiotics has resulted in the natural selection of bacteria that are resistant to many of these antibiotics, either because they have mutations in the antibiotic's target in the ribosome or because they have acquired a mechanism for excluding or inactivating the antibiotic. SEE ALSO CELL, EUKARYOTIC; RIBOZYME; RNA; TRANSLATION.

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Bibliography

- Frank, Joachim. "How the Ribosome Works." *American Scientist* 86 (1998): 428–439
- Garrett, Robert A., et al, eds. *The Ribosome: Structure, Function, Antibiotics, and Cellular Interactions*. Washington, DC: ASM Press, 2000
- Karp, Gerald. *Cell and Molecular Biology: Concepts and Experiments*, 3rd ed. New York: John Wiley & Sons, 2002.

Ribozyme

Ribozymes are RNA molecules that catalyze chemical reactions. Most biological processes do not happen spontaneously. For example, the cleavage of a molecule into two parts or the linkage of two molecules into one larger molecule requires **catalysts**, that is, helper molecules that make a reaction go faster. The majority of biological catalysts are proteins called enzymes. For many years scientists assumed that proteins alone had the structural complexity needed to serve as specific catalysts in cells, but around 1980 the research groups of Tom Cech and Sidney Altman independently discovered that some biological catalysts are made of RNA. These two scientists were honored with the Nobel Prize in chemistry in 1989 for their discovery.

Structure and Function

The RNA catalysts called ribozymes are found in the nucleus, mitochondria, and chloroplasts of **eukaryotic** organisms. Some viruses, including

catalysts substances that speed up a reaction without being consumed (e.g., enzyme)

eukaryotic describing an organism that has cells containing nuclei

several bacterial viruses, also have ribozymes. The ribozymes discovered to date can be grouped into different chemical types, but in all cases the RNA is associated with metal ions, such as magnesium (Mg^{2+}) or potassium (K^+), that play important roles during the catalysis. Almost all ribozymes are involved in processing RNA. They act either as molecular scissors to cleave precursor RNA chains (the chains that form the basis of a new RNA chain) or as “molecular staplers” that **ligate** two RNA molecules together. Although most ribozyme targets are RNA, there is now very strong evidence that the linkage of amino acids into proteins, which occurs at the ribosome during **translation**, is also catalyzed by RNA. Thus, the ribosomal RNA is itself also a ribozyme.

In some ribozyme-catalyzed reactions, the RNA cleavage and ligation processes are linked. In this case, an RNA chain is cleaved in two places and the middle piece (called the intron) is discarded, while the two flanking RNA pieces (called exons) are ligated together. This reaction is called splicing. Besides ribozyme-mediated splicing, which involves RNA alone, there are some splicing reactions that involve RNA-protein complexes. These complexes are called small nucleus ribonucleoprotein particles, abbreviated as snRNPs. This class of splicing is a very common feature of messenger RNA (mRNA) processing in “higher” eukaryotes such as humans. It is not yet known if snRNP-mediated splicing is catalyzed by the RNA components. Note also that some RNA splicing reactions are catalyzed by enzymes made of only protein.

Some precursor RNA molecules have a ribozyme built into their own intron, and this ribozyme is responsible for removal of the intron in which it is found. These are called self-splicing RNAs. After the splicing reaction is complete, the intron, including the ribozyme, is degraded. In these cases, each ribozyme works only once, unlike protein enzymes that catalyze a reaction repeatedly. Examples of self-spliced RNAs include the ribosomal RNAs of **ciliated protozoa** and certain mRNAs of yeast mitochondria.

Some RNA viruses, such as the hepatitis delta virus, also include a ribozyme as part of their inherited RNA molecule. During replication of the viral RNA, long strands containing repeats of the RNA **genome** (viral genetic information) are synthesized. The ribozyme then cleaves the long **multimeric** molecules into pieces that contain one genome copy, and fits that RNA piece into a virus particle.

Other ribozymes work on other RNA molecules. One ribozyme of this type is RNase P, which consists of one RNA chain and one or more proteins (depending on the organism). The catalytic mechanism of RNase P has been especially well-studied in bacteria. This ribozyme processes precursor transfer RNA (tRNA) by removing an extension from the 5-prime end, to create the 5-prime end of the “mature” tRNA (the two ends of an RNA molecule are chemically distinct and are called the 5-prime and 3-prime ends, referring to specific carbons in the sugar moiety of the terminal nucleotides). When the RNA molecule from bacterial RNase P is purified away from its protein, it can still cleave its precursor tRNA target, albeit at a very slow rate, proving that the RNA is the catalyst. Nevertheless, the protein(s) in RNase P also has important functions, such as maintenance of the proper conformation of the RNase P RNA and interaction with the precursor tRNA.

ligate join together

translation synthesis of protein using mRNA code

ciliated protozoa single-celled organism possessing cilia, short hair-like extensions of the cell membrane

genome the total genetic material in a cell or organism

multimeric describes a structure composed of many similar parts



Relics of an “RNA World”

Many biologists hypothesize that ribozymes are vestiges of an ancient, prebiotic world that predated the evolution of proteins. In this “RNA world,” RNAs were the catalysts of such functions as replication, cleavage, and ligation of RNA molecules. Proteins are hypothesized to have evolved later, and as they evolved they took over functions previously performed by RNA molecules. This may have happened because proteins are more versatile and efficient in their catalytic functions.

In today’s world, most processing of precursor tRNA is performed by the ribozyme RNase P, as described above, but in some chloroplasts, this function is performed by a protein that apparently contains no RNA. This may be an example of the evolution of protein enzymes that replace ribozymes.

Intensive studies of ribozymes have provided rules for how they recognize their targets. Based on these rules, it has been possible to alter ribozymes to recognize and cleave new targets in RNA molecules that are normally not subject to ribozyme cleavage. These results raise the exciting possibility of using ribozymes for human therapy. For example, the abundance of disease-causing RNA molecules such as HIV, the cause of AIDS, could be reduced with artificial ribozymes. Considerable success has been achieved in testing these ribozymes in model cells. However, the biggest question remaining to be solved is how these potential “disease-fighting” ribozymes can be introduced into a patient and taken up by the appropriate cells. **SEE ALSO** EVOLUTION; MOLECULAR; PROTEINS; RNA; RNA PROCESSING.

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Bibliography

Cech, T. R. “RNA as an Enzyme.” *Scientific American* 255 (1986): 64–75.

Karp, Gerald. *Cell and Molecular Biology*, 3rd ed. New York: John Wiley & Sons, 2002.

RNA

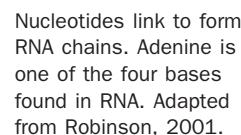
Ribonucleic acid (RNA) molecules, which are linear chains (or polymers) of ribonucleotides, perform a number of critical functions. Many of these functions are related to protein synthesis. Some RNA molecules bring genetic information from a cell’s chromosomes to its **ribosomes**, where proteins are assembled. Others help ribosomes translate genetic information to assemble specific sequences of amino acids.

Molecular Structure

Ribonucleotides, the building blocks of RNA, are molecules that consist of a nitrogen-containing base, a phosphate group, and ribose, a five-carbon sugar. The nitrogen-containing base may be adenine, cytosine, guanine, or uracil. These four bases are abbreviated as A, C, G, and U.

RNA is similar to deoxyribonucleic acid (DNA), another class of nucleic acid. However, DNA nucleotides contain deoxyribose, not ribose, and they use the nitrogen-containing base thymine (T), not uracil, along with adenine, cytosine, and guanine.

ribosomes protein-RNA complexes at which protein synthesis occurs



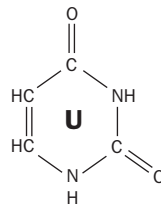
Tertiary structures usually involve interactions between nucleotides that are distant from each other along an RNA strand. Such interactions may arise from hydrogen bonding between bases, as in regular Watson-Crick base pairing, or from interactions among other chemical groups in the nucleotides. Some RNA molecules, such as ribosomal RNA (rRNA)

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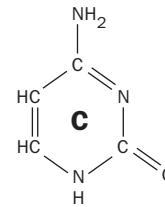
The molecular structures of the four RNA bases. Adapted from Robinson, 2001.

BASES

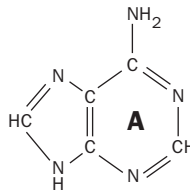
The bases are nitrogen-containing ring compounds, either purines or pyrimidines.



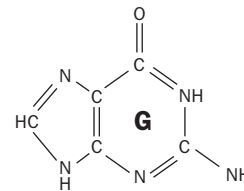
uracil



cytosine



adenine



guanine

and transfer RNA (tRNA), have structures that are very complex. In structure they resemble proteins more than they do DNA.

To understand the function of a given RNA molecule, scientists often need to know its structure. There are three general strategies for analyzing RNA structure. First, using the relatively simple base-pairing rules for RNA and the basic principles of **thermodynamics**, computers can be used to predict secondary RNA structure, although not always with complete success.

Second, researchers can analyze RNA molecules from various organisms and compare those molecules that have the same function. Even when the nucleotide sequences vary between species, important structures are usually preserved.

Third, the structure of an RNA molecule can be determined experimentally, using **enzymes** to cut it or chemicals to modify it. Some enzymes and chemicals cut or modify only nonpaired, single-stranded portions of the RNA molecule, allowing researchers to identify double-stranded regions by examining which ones remain uncut and unmodified.

Despite the usefulness of each of these methods, none can provide a complete and accurate three-dimensional structure. A more complete determination of structure can be achieved by the biophysical methods of X-ray crystallography and nuclear magnetic resonance.

Synthesis

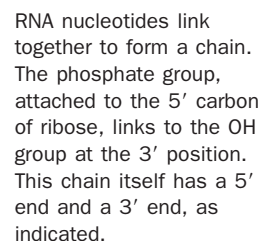
RNA molecules are synthesized by enzymes known as RNA polymerases in a process called transcription. Usually, one strand of a double-stranded DNA molecule is used as a **template** for the RNA. The order of ribonucleotides that are assembled to form the RNA molecule is determined by the order of the deoxyribonucleotides in the DNA strand. The genetic information

thermodynamics

process of energy transfers during reactions, or the study of these processes

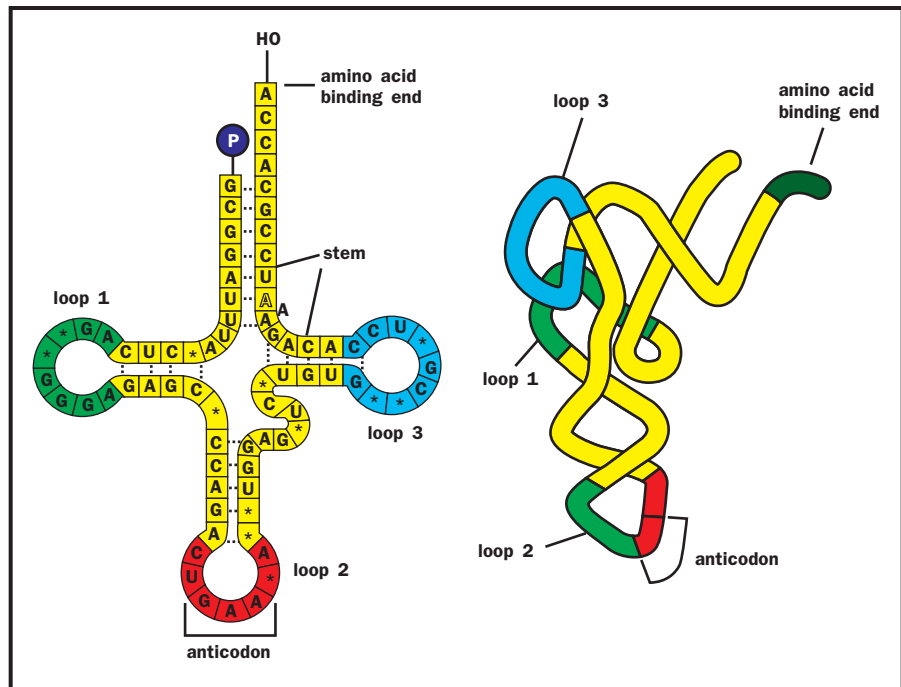
enzymes proteins that control a reaction in a cell

template a master copy



Often, when RNA molecules are synthesized, they are in a form that prevents them from carrying out their function. To become functional, they must undergo processing, which can involve removing segments of the strands or modifying specific nucleotides. The link between a base and a ribose may be altered, or extra chemical groups may be added to the bases or ribose molecules. Many RNA molecules are associated with proteins during or after their synthesis. Together, the RNA and protein are referred to as RNA-protein particles (RNPs).

The clover-leaf shape (left) is characteristic of transfer RNA. The L shaped figure (right) is the more compacted form the tRNA adopts in the cell. The asterisks within the RNA sequence indicate modified RNA nucleotides that occur in tRNA only. The anticodon binds with a complementary sequence on messenger RNA. Adapted from Robinson, 2001.



eukaryotes organisms with cells possessing a nucleus

cytoplasm the material in a cell, excluding the nucleus

prokaryotes single-celled organisms without a nucleus

In **eukaryotes**, RNA that is encoded by nuclear chromosomes is synthesized in the nucleus. The processing and assembly of many small RNA molecules in higher eukaryotes is accomplished in Cajal bodies, which are coiled structures in the nucleus that were identified more than 100 years ago but that have begun to be investigated in detail only recently. The synthesis of those RNA molecules that are components of ribosomes occurs in the nucleolus, a part of the nucleus. RNA synthesis and processing also occurs in the mitochondria and chloroplasts, when the RNA will be used in those organelles.

After being processed and assembled, RNPs either remain in the nucleus or are exported to the **cytoplasm** through the nuclear pores. Some are also exported and modified in the cytoplasm and then imported back into the nucleus. In **prokaryotes**, where there is no nucleus, the synthesis and processing of RNA, as well as the assembly of RNPs, occurs in the cytoplasm.

Function

Almost all types of RNA play a role in translation, which is the process of protein synthesis. Translation requires three types of RNA: messenger RNA (mRNA), which ranges in length from a few hundred to many thousands of nucleotides; tRNA, which is 75 to 85 nucleotides long; and rRNA, which is 1,500 to 4,000 nucleotides long.

Molecules of mRNA, each of which contains a copy of at least one gene, are the intermediates between DNA and protein. These mRNA molecules bring the genetic code from the DNA, which is in the nucleus, to ribosomes, which are in the cytoplasm. They attach to the ribosomes and determine the order in which amino acids are assembled to synthesize a protein. Of

the three types of RNA required for translation, mRNA molecules have the simplest structure.

Next, tRNA molecules function as adapters that help translate the nucleotide sequences in mRNA into amino acid sequences, so specific proteins can be constructed. There are many different types of tRNA, each of which is capable of binding to one of the twenty amino acids that are the building blocks of proteins.

Finally, rRNA molecules, which account for most of a ribosome's mass, are, according to recent experiments, the part of the ribosome responsible for linking amino acids into a growing protein chain. Ribosomes, the organelles that assemble a particular sequence of amino acids to form proteins, contain three or four different molecules of rRNA, along with at least fifty different proteins.

Both rRNA and tRNA are stable forms of RNA that last through several cell divisions. In contrast, mRNA is normally unstable, with a lifetime that can be as short as a few minutes. This instability has probably evolved because it lets cells quickly stop synthesizing proteins that are no longer needed. In some cases, enzymes called ribonucleases (RNases) actively degrade certain mRNA molecules. For example, mRNA that encodes a particular protein regulating the **cell cycle** is degraded when the protein has carried out its function.

In certain cells, mRNA can exist in a stable form for decades. When egg cells are formed, for example, some of the mRNA in the cells is associated with “storage proteins” and lasts until after the eggs are fertilized. During embryonic development, this maternal mRNA becomes activated for translation and associates with translating ribosomes. It usually decays after it has been used to produce a certain amount of protein.

Less Common Types of RNA

Several types of less abundant, small RNA molecules perform essential functions in both the nucleus and the cytoplasm. All organisms contain cytoplasmic RNPs that are involved in exporting proteins from cells. During the synthesis of proteins that are destined to be exported, the ribosome and mRNA associate with an “export-RNP,” which helps them dock at an export pore in the cell membrane. As it formed, the protein is threaded through the membrane to the outside of the cell. In eukaryotes, this same strategy is used to transport proteins into the **endoplasmic reticulum**, where some newly synthesized proteins are sorted and modified.

RNase P is another RNP found in all forms of life. This RNA-containing enzyme helps turn precursor tRNA into mature tRNA molecules. It does so by cleaving a section off the 5' end of the precursor molecules.

Small nucleolar RNAs, which are known as snoRNAs and which are found in the nucleoli of eukaryotes and in **Archaea**, are required for the processing of precursor rRNA. During the assembly of new ribosomes, snoRNAs help remove regions of the precursor molecules and modify specific nucleotides.

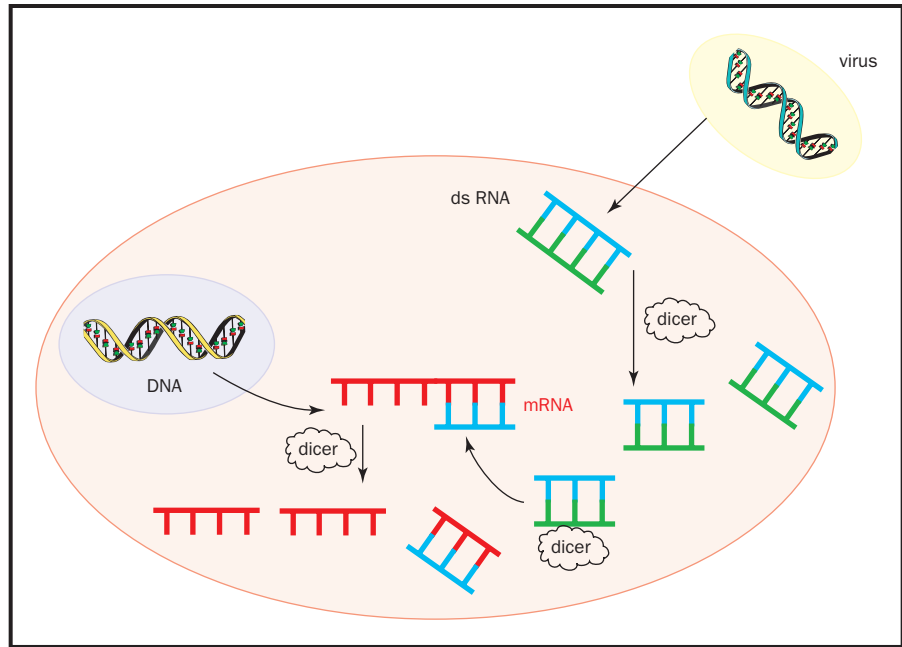
Often, mRNA molecules in eukaryotes and in Archaea contain sequences that do not code for amino acids. These sequences, called introns, must be

cell cycle sequence of growth, replication and division that produces new cells

endoplasmic reticulum network of membranes within the cell

Archaea one of three domains of life, a type of cell without a nucleus

Proposed mechanism for RNA interference. Double-stranded RNA from a virus is recognized and degraded by the enzyme dicer. These fragments can then pair up with complementary host messenger RNA (mRNA), causing further degradation and gene silencing.



spliced out before translation begins. In eukaryotes, small nuclear RNAs (snRNAs) in the nucleus remove these introns. Once the introns are removed, the mature mRNA molecules are exported, through nuclear pores, into the cytoplasm, where they associate with ribosomes for translation.

Some viral genomes consist of single-stranded or double-stranded RNA, not DNA. Examples are found among both prokaryotic and eukaryotic viruses and include HIV, as well as viruses causing some forms of cancer. SEE ALSO EVOLUTION, MOLECULAR; NUCLEOTIDE; RIBOSOME; RNA POLYMERASES; RNA PROCESSING; TRANSCRIPTION; TRANSLATION.

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Bibliography

- Lodish, Harvey, et al. *Molecular Cell Biology*, 4th ed. New York: W. H. Freeman, 2000.
- Meili, M., B. Albert-Fournier, and M. C. Maurel. "Recent Findings in the Modern RNA World." *International Microbiology* 4 (2001): 5–11.
- Robinson, Richard. *Biology*. Farmington Hills, MI: Macmillan Reference USA, 2002.
- Storz, G. "An Expanding Universe of Non-coding RNAs." *Science* 296 (2002): 1260–1263.

RNA Interference

translation synthesis of protein using mRNA code

RNA interference is a process in which **translation** of some of a cell's messenger RNA (mRNA) sequences is prevented, because of the presence of (and consequent destruction of) matching double-stranded RNA sequences. RNA interference is believed to protect the cell against viruses and other threats. "Interference" refers to the interruption of the cell's translation of its own mRNA. RNA interference is also called posttranscriptional gene silencing, since its effect on gene expression occurs after the creation of the mRNA during transcription.

RNA interference has been found in plants, fungi, and a variety of animals, including the roundworm (*Caenorhabditis elegans*), fruit fly (*Drosophila melanogaster*), zebrafish, and mouse. It is believed to be an ancient form of defense. It may also explain some or most of the gene-silencing effect of “antisense” RNA, as discussed elsewhere in this encyclopedia.

Dicing up dsRNA

Under most circumstances, RNA in a cell is present as a single-stranded molecule only. For instance, mRNA is created in the cell nucleus and transported to the ribosomes in the cytoplasm as a single strand. Double-stranded RNA (dsRNA), in which two **complementary** strands pair up, is normally present only in circumstances that pose a threat to the cell. This can occur when a dsRNA virus infects the cell, or from infection by some other viruses whose genomes are temporarily copied into dsRNA. It also occurs when certain types of transposable genetic elements (**transposons**) copy themselves in preparation for reinserting elsewhere in the cell’s genome. Though the RNA copies are single-stranded, most transposons have sequences at their ends that, when transcribed into RNA, can fold back on themselves to form dsRNA.

When a cell detects dsRNA, it uses a **nuclease** enzyme to cut it into small fragments, twenty-one to twenty-three nucleotides long (the *Drosophila* enzyme is whimsically but accurately named “dicer”). This inactivates the RNA, so that it cannot be used to carry out the viral replication cycle or be reinserted into the genome (in the case of a transposon), thus protecting the cell from its harmful effects.

Interference

Degradation of the dsRNA is not the end of the process, however. The presence of these fragments also prevents the expression of mRNA containing the same sequences. That is, if the host cell has used its own gene to create a single-stranded mRNA, and that mRNA is present in the cytoplasm along with dsRNA fragments with matching nucleotide sequences, the mRNA will be degraded, and the protein it codes for will not be made. This is the “interference” that gives the phenomenon its name. Indeed, it was this process that led to the discovery of RNA interference: Scientists found that introducing double-stranded RNA reduced, rather than increased, production of the encoded protein.

Note that not all mRNA activity in the cell is suppressed: Only those mRNAs of similar sequence are targeted. This provides a clue to the mechanism of suppression. Experiments have shown that dicer targets mRNA by using the dsRNA fragments themselves as guides. While the details are not yet clear, it is believed that one side of the dsRNA is matched with the complementary mRNA sequence, making a new dsRNA, which is itself then degraded. This mechanism also allows the process to be self-sustaining, as each new round creates new fragments that can target any new mRNA.

The recognition process also lends further credence to the belief that RNA interference is a protective mechanism. By targeting only mRNA sequences previously identified as double-stranded (and therefore dangerous),

complementary matching opposite, like hand and glove

transposons genetic elements that move within the genome

nuclease enzyme that cuts DNA or RNA



a cell can avoid creating proteins that may be derived from viruses, albeit at the risk of turning off one or more of its own genes at the same time.

Research Uses of RNA Interference

Because of its ability to turn off individual gene expression, RNA interference provides a remarkably precise tool for studying the effects of individual genes. There are several ways to deliver dsRNA to cells. It can be injected into a single cell or placed in a viral chromosome that infects the cells being studied. Roundworms will absorb dsRNA if they are soaked in a solution containing it, or if they eat bacteria that contain it.

RNA interference has several advantages over the alternative way of “knocking out” a gene, called gene targeting. Unlike gene targeting, administration of dsRNA does not require long and laborious breeding of the target organism carrying the knockout. Even more importantly, the dsRNA knockout is temporary and can be induced at any stage of the life cycle, rather than exerting its effect throughout life, as with gene targeting. This allows short-term studies of gene effects, a feature particularly valuable for studying development, for instance.

New Developments in dsRNA

Recent research has also shown that a class of similar dsRNA fragments, called small temporal RNAs, play important roles in development in the roundworm, fruit fly, and other animals. Although little is so far known about them, these fragments are made by dicer from the cell’s own RNA as a normal part of the developmental process and appear to help control **gene expression**. This is an exception to the statement that the presence of dsRNA signals a threat to the cell; how these are distinguished from threatening dsRNA is not yet known. **SEE ALSO** ANTISENSE NUCLEOTIDES; FRUIT FLY: *DROSOPHILA*; NUCLEASES; RNA; POST-TRANSLATIONAL CONTROL; RNA PROCESSING; ROUNDWORM: *CAENORHABDITIS ELEGANS*; TRANSPOSABLE GENETIC ELEMENTS; VIRUS.

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Bibliography

Ambros, Victor. “Development: Dicing up RNAs.” *Science* 293 (2001): 811–813.

Gura, Trisha. “A Silence that Speaks Volumes.” *Nature* 404 (2000): 804–808.

Lin, Rueyling, and Leon Avery. “RNA Interference. Policing Rogue Genes.” *Nature* 402 (1999): 128–129.

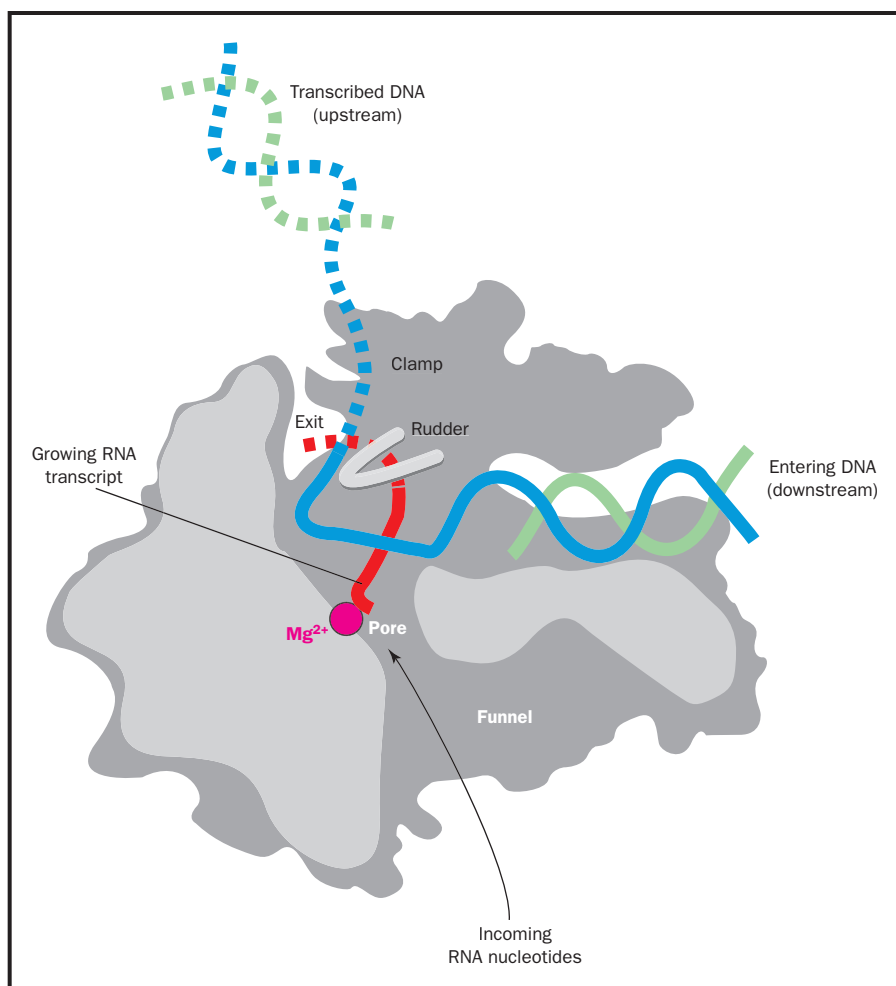
RNA Polymerases

RNA polymerases are enzyme complexes that synthesize RNA molecules using DNA as a **template**, in the process known as transcription. The RNAs created by transcription are either used as is (as ribosomal RNAs, transfer RNAs, or other types), or serve to guide the **synthesis** of a protein (as messenger RNAs). The word “polymerase” derives from “-ase,” a suffix indicating an enzyme, and “polymer,” meaning a large molecule composed of many similar parts, in this case the RNA nucleotides A, U, C, and G (abbreviations for adenine, uracil, cytosine, and guanine).

gene expression use of a gene to create the corresponding protein

template a master copy

synthesis creation



RNA polymerase use one side of the DNA double helix (blue) to assemble RNA nucleotides into an RNA transcript. The internal structure of the polymerase, shown in cut-away view, helps separate the DNA strands. A magnesium ion (Mg^{2+}) helps catalyze the addition of RNA nucleotides into the growing chain. Adapted from <http://www.euchromatin.org/1844-1-med.gif>.



Types of Polymerases

Prokaryotic organisms (Eubacteria and Archaea) have only one type of RNA polymerase. **Eukaryotic** organisms (animals, plants, fungi, and protists) have three types, called pol I, II, and III, and each transcribes a different set of genes. Pol I synthesizes RNA for the large subunit of the ribosome (the protein-making machinery of the cell), and one piece of RNA for the small subunit. Pol II creates messenger RNAs, which provide a template for protein synthesis. Pol II also creates numerous small nuclear RNAs (snRNAs), which modify RNAs after they are formed. Pol III synthesizes transfer RNAs (tRNAs), the RNA for the small subunit of the ribosome, and other snRNAs.

The three eukaryotic polymerases can be distinguished in the laboratory by the degree to which they are inhibited by the alpha-amanitin poison from the mushroom *Amanita phalloides*. Pol I is completely resistant to its effects, pol III is moderately sensitive, and pol II is highly sensitive. (The reason this poison is so deadly is precisely because it inhibits RNA polymerase.)

Each eukaryotic RNA polymerase is composed of a dozen or more subunits. Some of these are small and unique to each type, but the two largest subunits are similar among all three polymerases, and similar as well to the two largest prokaryotic subunits. This is clear evidence that all of them

prokaryotic to come

eukaryotic describing an organism that has cells containing nuclei

transcription factors
proteins that increase
the rate of gene tran-
scription

catalyze aid in the
reaction of

evolved from the same original polymerase. These shared subunits are thought to form the functional core of the polymerases, while the smaller subunits may provide the gene specificity of each type.

Transcription

Transcription begins when RNA polymerase binds to the DNA double helix. This occurs at a site just “upstream” of the gene to be transcribed, called the promoter site. In eukaryotes, RNA polymerase is directed to the promoter site by **transcription factors**, proteins that bind to the DNA and provide a docking site for attachment of the polymerase enzyme. Once RNA polymerase binds to the DNA at the promoter, transcription can begin.

During transcription, the polymerase unwinds a portion of the double-stranded DNA, exposing the DNA template strand that will be copied into RNA. Individual RNA nucleotides enter the enzyme complex, and are paired with the DNA. C pairs with G, T (on DNA) pairs with A, and A (on DNA) pairs with U. Nine DNA-RNA nucleotide pairs exist within the polymerase molecule at any one time. After each new RNA nucleotide is paired, it is linked to the preceding RNA nucleotide, forming a growing strand of polymerized RNA called the primary transcript. This stage of transcription is called elongation.

Recent X-ray analysis of RNA polymerase has revealed important structural details that help explain the precise mechanism of transcription. Double-helical DNA enters a long cleft in the surface of the enzyme, and is held in place by a large flexible portion of the enzyme termed the “clamp.” Within the cleft, the DNA is separated and RNA is paired to it. A magnesium ion, sitting at the critical point where RNA nucleotides are added to the primary transcript, is thought to help **catalyze** this reaction. An internal barrier forces a bend in the growing DNA-RNA duplex, exposing the RNA end for addition of the incoming nucleotide. A short protein extension, termed the “rudder,” helps to separate the RNA from the DNA, and the two exit the polymerase along separate paths.

The average maximum rate of elongation in bacteria is 5 to 10 nucleotides per second. However, during transcription, the polymerase enzyme may pause for seconds to minutes. These pauses are thought to be part of a regulatory mechanism. Transcription continues until RNA polymerase reaches a special DNA sequence called the termination sequence, at which point it detaches from the DNA. SEE ALSO NUCLEOTIDE; RNA; RNA PROCESSING; TRANSCRIPTION; TRANSCRIPTION FACTORS.

Richard Robinson

Bibliography

Alberts, Bruce, et al. *Molecular Biology of the Cell*, 3rd ed. New York: Garland Publishing, 1994.

Klug, Aaron. “A Marvellous Machine for Making Messages.” *Science* 292, no. 5523 (2001): 1844–1846.

White, Robert J. *Gene Transcription: Mechanisms and Control*. Oxford: Blackwell Science, 2001.

Internet Resource

“Transcribed DNA.” Euchromatin Forums. <<http://www.euchromatin.org/1844-1-med.gif>>.

RNA Processing

RNA serves a multitude of functions within cells. These functions are primarily involved in converting the genetic information contained in a cell's DNA into the proteins that determine the cell's structure and function. All RNAs are originally transcribed from DNA by RNA polymerases, which are specialized **enzyme** complexes, but most RNAs must be further modified or processed before they can carry out their roles. Thus, RNA processing refers to any modification made to RNA between its transcription and its final function in the cell. These processing steps include the removal of extra sections of RNA, specific modifications of RNA bases, and modifications of the ends of the RNA.

Types of RNA

There are different types of RNA, each of which plays a specific role, including specifying the amino acid sequence of proteins (performed by messenger RNAs, or mRNAs), organizing and catalyzing the synthesis of proteins (ribosomal RNAs or rRNAs), translating codons in the mRNA into amino acids (transfer RNAs or tRNAs) and directing many of the RNA processing steps (performed by small RNAs in the nucleus, called snRNAs and snoRNAs).

All of these types of RNAs begin as primary transcripts copied from DNA by one of the RNA polymerases. One of the features that separates **eukaryotes** and **prokaryotes** is that eukaryotes isolate their DNA inside a nucleus while protein synthesis takes place in the cytoplasm. This separates the processes of transcription and translation in space and time. Prokaryotes, which lack a nucleus, can translate an mRNA as soon as it is transcribed by RNA polymerase. As a consequence, there is very little processing of prokaryotic mRNAs. By contrast, in eukaryotic cells many processing steps occur between mRNA transcription and translation. Unlike the case of mRNAs, both eukaryotes and prokaryotes process their rRNAs and tRNAs in broadly similar ways.

Types of RNA Processing

There are three main types of RNA processing events: trimming one or both of the ends of the primary transcript to the mature RNA length; removing internal RNA sequences by a process called RNA splicing; and modifying RNA nucleotides either at the ends of an RNA or within the body of the RNA. We will briefly examine each of these and then discuss how they are applied to the various types of cellular RNAs.

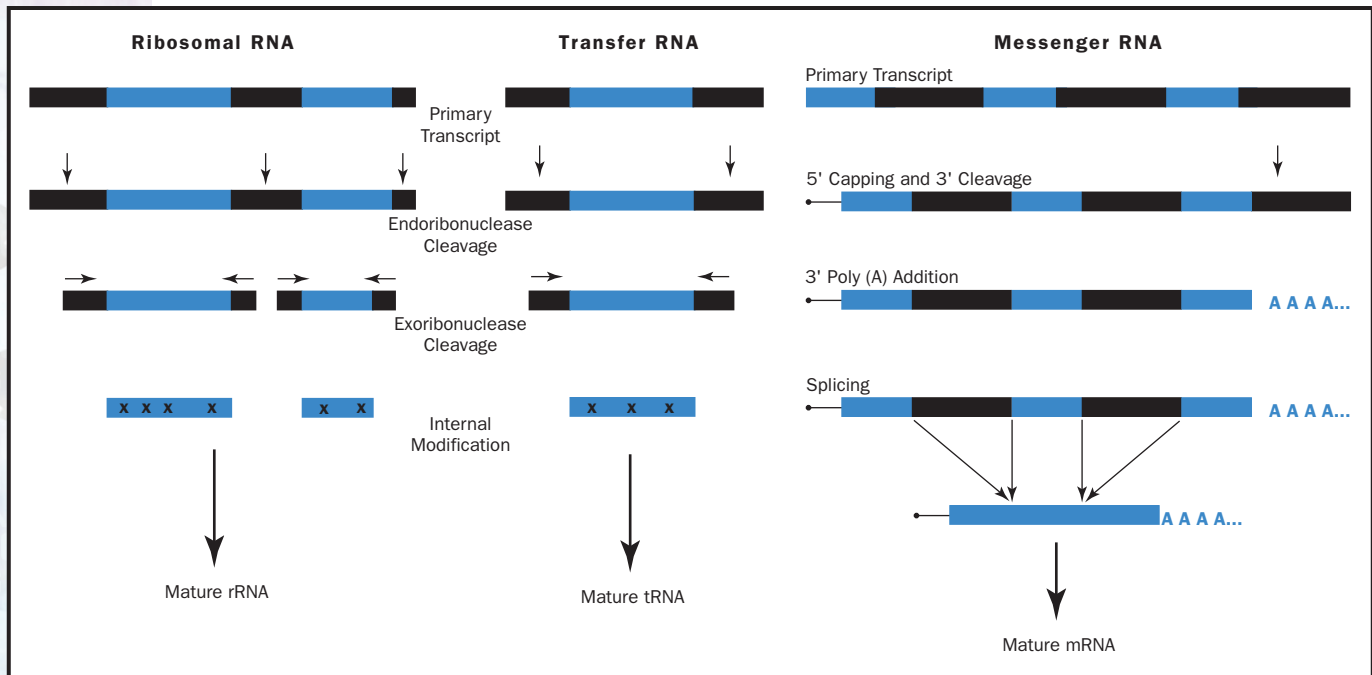
Almost all RNAs have extra sequences at one or both ends of the primary transcripts that must be removed. The removal of individual nucleotides from the ends of the RNA strand is carried out by any of several ribonucleases (enzymes that cut RNA), called exoribonucleases. An entire section of RNA sequence can be removed by cleavage in the middle of an RNA strand. The enzymes responsible for the cleavage in this location are called endoribonucleases. Each of these ribonucleases is targeted so that it only cleaves particular RNAs at particular places.

enzyme a protein that controls a reaction in a cell

eukaryotes organisms with cells possessing a nucleus

prokaryotes single-celled organisms without a nucleus





The principal steps in processing various types of RNAs in eukaryotic nuclei. RNA sequences that are in the mature RNAs are indicated in blue while RNA sequences that are removed during processing are in black. Small "x's" indicate sites of internal modifications of RNAs. The various steps do not necessarily occur in the order shown.

RNA splicing is similar to trimming in that it removes extra RNA sequences, but it is different because the sequence is removed from the middle of an RNA and the two flanking pieces are joined together again (see figure). The part of the RNA that is removed is called an intron, whereas the two pieces that are joined together, or spliced, are called exons. Just as with the cleavage enzymes, the splicing machinery recognizes particular sites within the RNA, in this case the junctions between exons and introns, and cleaves and rejoins the RNA at those positions.

Modification of RNA nucleotides can occur at the ends of an RNA molecule or at internal positions. Modification of the ends can protect the RNA from degradation by exoribonucleases and can also act as a signal to guide the transport of the molecule to a particular subcellular compartment. Some internal modifications, particularly of tRNAs and rRNAs, are necessary for these RNAs to carry out their functions in protein synthesis. Some internal modifications of mRNAs change the sequence of the message and so change the amino acid sequence of the protein coded for by the mRNA. This process is called RNA editing. As with the other types of RNA processing, the enzymes that modify RNAs are directed to specific sites on the RNA.

Processing of Various Classes of RNAs

Ribosomal RNAs are synthesized as long primary transcripts that contain several different rRNAs separated by spacer regions (see figure). The individual rRNAs are cut apart by endoribonucleases that cleave within the spacer regions. Other enzymes then trim the ends to their final length. Ribosomal RNAs are also modified at many specific sites within the RNA. Ribosomal RNA synthesis and processing occurs in a special structure within the nucleus called the **nucleolus**. The mature rRNAs bind to ribosomal proteins within the nucleolus and the assembled ribosomes are then transported to the **cytoplasm** to carry out protein synthesis.

nucleolus portion of the nucleus in which ribosomes are made

cytoplasm the material in a cell, excluding the nucleus

Transfer RNAs are transcribed individually from tRNA genes. The primary transcripts are trimmed at both the 5' and 3' ("five prime," or "upstream" and "three prime," or "downstream") ends, and several modifications are made to internal bases. Many eukaryotic tRNAs also contain an intron, which must be removed by RNA splicing. The finished tRNAs are then transported from the nucleus to the cytoplasm.

Messenger RNAs are transcribed individually from their genes as very long primary transcripts. This is because most eukaryotic genes are divided into many exons separated by introns. Genes may contain from zero to more than sixty introns, with a typical gene having around ten. Introns are spliced out of primary RNA transcripts by a large structure called the **spliceosome**. The spliceosome does not move along the RNA but is assembled around each intron where it cuts and joins the RNA to remove the intron and connect the exons. This must be done many times on a typical primary transcript to produce the mature mRNA.

In addition to removal of the introns, the mRNA is modified at the 5' end by the addition of a special "cap" structure that is later recognized by the translation machinery. The mRNA is also trimmed at the 3' end and several hundred adenosine nucleotides are added. This modification, which is called either polyadenylation or poly (A) addition, helps stabilize the 3' end against degradation and is also recognized by the translation machinery. Finally, the processed mature mRNA is transported from the nucleus to the cytoplasm.

Some RNAs, called small nuclear RNAs (snRNAs) and small nucleolar RNAs (snoRNAs), are processed in the nucleus and are themselves part of the RNA processing systems in the nucleus. Most snRNAs are involved in mRNA splicing, while most snoRNAs are involved in rRNA cleavage and modification.

RNA Processing and the Human Genome

The fact that most human genes are composed of many exons has some important consequences for the expression of genetic information. First, we now know that many genes are spliced in more than one way, a phenomenon known as alternative splicing. For example, some types of cells might leave out an exon from the final mRNA that is left in by other types of cells, giving it a slightly different function. This means that a single gene can code for more than one protein. Some complicated genes appear to be spliced to give hundreds of alternative forms. Alternative splicing, therefore, can increase the coding capacity of the genome without increasing the number of genes.

A second consequence of the exon/intron gene structure is that many human gene mutations affect the splicing pattern of that gene. For example, a mutation in the sequence at an intron/exon junction that is recognized by the spliceosome can cause the junction to be ignored. This causes splicing to occur to the next exon in line, leaving out the exon next to the mutation. This is called exon skipping and it usually results in an mRNA that codes for a nonfunctional protein. Exon skipping and other errors in splicing are seen in many human genetic diseases. **SEE ALSO ALTERNATIVE SPLICING; NUCLEOTIDE; NUCLEUS; RIBOSOME; RNA; RNA POLYMERASES.**

Richard A. Padgett

spliceosome RNA-protein complex that removes introns from RNA transcripts



Bibliography

Lewin, Benjamin. *Genes VII*. Oxford, U.K.: Oxford University Press, 2000.

Lodish, Harvey, et al. *Molecular Cell Biology*, 4th ed. New York: W. H. Freeman, 2000.

Rodent Models

Rodents play an important role in biology and medicine. Since the mouse and rat share many biological characteristics with humans, they are commonly used as model organisms for understanding disease processes and testing treatments. Moreover, it is relatively easy to experimentally manipulate the genetic composition of mice and rats and, thereby, to model human genetic disorders in these animals.

Large numbers of mice can be raised quickly and relatively cost-effectively. Mice have short life spans and relatively short generation times. This makes the mouse an extremely useful experimental system for studying the genetics and biology of human disease. Rats are slightly more similar to humans than are mice, but rats are larger, have a longer generation time, are less well-characterized genetically, and are not as easily genetically manipulated. So while rats are useful in the study of some aspects of biology, mice are now much more widely used in molecular genetic research.

Many features of human biology at the cellular and molecular levels are shared with a wide variety of organisms. As mammals, the mouse and rat are highly related to humans, with similar genes, biochemical pathways, organs, and physiology.

The **genomes** of both human and mouse have now been sequenced, and they show striking similarity. Estimates of the total number of genes in both genomes are very comparable, ranging from 30,000 to 40,000. It is thought that only about 1 percent of human genes are unique and do not have a mouse counterpart. In addition, mouse genes are on average approximately 85 percent similar in sequence to their human counterparts. Thus, mice and humans are very highly related at the genetic level.

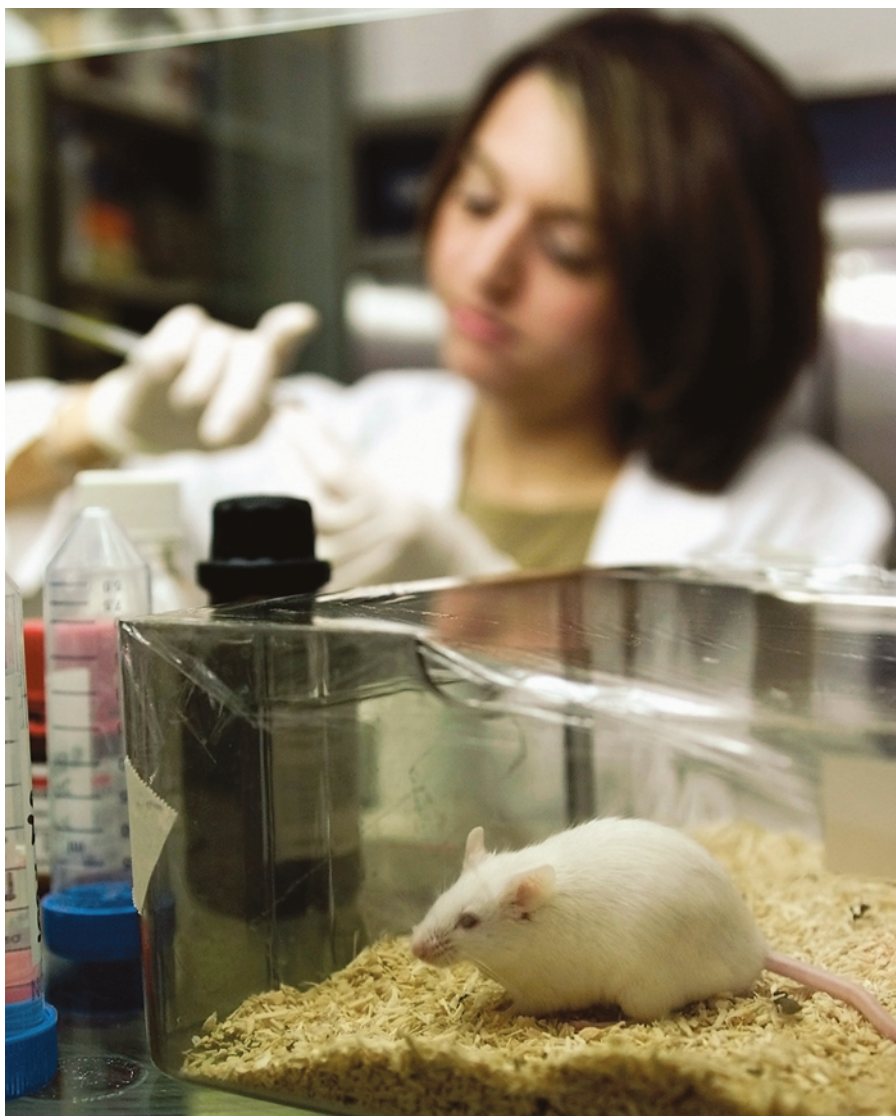
Before the advent of genetic modification techniques in the twentieth century, collecting mouse strains with unusual characteristics was a pastime of amateur enthusiasts. These variant strains, often showing physical characteristics such as unusual fur or tails, arose as a result of spontaneous **mutations**. The discovery that radiation and chemicals could increase the rate of mutation led to the development of laboratory mouse stocks, which are a useful source of genetic variation. Over 1,000 spontaneous or radiation-induced mutations have been documented, and many have been the starting point for the study of relevant human diseases and biology. For example, the cloning of the obese and diabetic mouse mutants identified the **hormone** leptin and its receptor, respectively, and opened up an entirely new area of research into the control of various physiological processes, including the control of body weight.

The importance of mice in genetic studies was first recognized in the related areas of immunology and cancer research, for which a mammalian model of human diseases was essential. Mice have been used in cancer research for almost a century, starting with the breeding of mouse strains

genomes total genetic material in a cell or organism

mutations changes in DNA sequences

hormones molecules released by one cell to influence another



Humans share many biological characters with rats and mice, which makes the rodents valuable models in the laboratory for researchers who are looking to find cures for genetic diseases.

susceptible to particular tumor types and, subsequently, in the induction of tumors by chemical compounds. These experiments were critical for demonstrating that the development of cancer is a multistep process, consisting of a series of genetic changes.

Although it has long been obvious that many other aspects of human biology and development can be studied using mouse models, until recently the methods for doing so did not exist. Mouse models moved to the forefront of modern biomedical research with the emergence of **recombinant DNA** technology, thirty years ago, and the pace has been accelerating ever since. The advent of this technology eventually culminated in the publication of the complete human genome sequence in 2001, and by 2002 the sequence and structure of almost every mouse gene will also be known. In combination with advances in DNA cloning, methods were developed for adding genes to mice (called transgenes) and removing genes (called gene targeting). These introduced genetic changes allow researchers to model human diseases by altering specific genes that are altered in particular human disease states.

recombinant DNA DNA formed by combining segments of DNA, usually from different types of organisms



For example, the mouse counterpart of *atm*, the gene that causes the human disease ataxia telangiectasia, was identified. In humans the symptoms of this disease include movement problems, abnormal blood vessels, immune defects, and a predisposition to cancer. The mouse *atm* gene was specifically disrupted by gene targeting, and mice lacking this gene exhibit a range of problems similar to those exhibited by the human patients. This mouse model is useful not only for studying and understanding the function of the gene in humans and its role in the disease process, but also for testing potential therapies.

Now that the sequencing of the human and mouse genes is completed, there is a large gap between the number of genes that have been identified and our understanding of how those genes function. The next challenge is to start to determine how those genes function in normal development and how these genes are disrupted in various disease conditions. Arguably, mouse models will become even more important during this phase of the analysis of the genome, as mice are amenable to a systematic genome-wide mutation of genes, which will allow the function of large numbers of genes to be determined. **SEE ALSO** GENE TARGETING; MODEL ORGANISMS; RECOMBINANT DNA; TRANSGENIC ANIMALS.

Seth G. N. Grant and Douglas J. C. Strathdee

Bibliography

- Jackson, Ian J., and Catherine M. Abbot, eds. *Mouse Genetics and Transgenics: A Practical Approach*. New York: Oxford University Press, 2000.
- Lyon, Mary F., Sohaila Rastan, and S. D. M. Brown. *Genetic Variants and Strains of the Laboratory Mouse*, 3rd ed. New York: Oxford University Press, 1996.
- Rossant, Janet, and Patrick P. L. Tam, eds. *Mouse Development: Patterning, Morphogenesis, and Organogenesis*. San Diego: Academic Press, 2001.

Roundworm: *Caenorhabditis elegans*

nematode worm of the Nematoda phylum, many of which are parasitic

Caenorhabditis elegans is a nonparasitic **nematode** that normally lives in the soil. Although studied since the 1800s, the modern use of *C. elegans* as a model system dates to the mid-1960s. By the start of that decade, scientists thought that the “classical” problems in molecular biology were about to be solved. They began to search for a model system that would support the new challenges presented by research on multicellular organisms, particularly in developmental biology and neurology. In 1965 Sydney Brenner proposed what he thought was the ideal model organism, one that provided the best compromise between biological complexity and ease of manipulation: *C. elegans*.

Useful Characteristics

C. elegans is hermaphroditic, meaning that almost all of the 300 progeny produced in a single clutch of eggs are females, capable of self-fertilization. In essence, *C. elegans* clones itself. One or two progeny are **morphologically** distinct males that can be collected and used for standard genetic studies. This permits rapid chromosomal mapping of the many **mutations** that occur in any one of the organism’s six chromosomes. The *C. elegans* genome was sequenced in 1998, predicting 19,099 genes. Currently, 3,000 of these are considered

morphologically related to shape and form

mutations changes in DNA sequences



Micrograph of a *Caenorhabditis elegans* nematode, magnified fifty times.

“essential” genes, 900 of which had already been identified through chromosomal mapping.

In the laboratory, *C. elegans* is easily cultivated on simple growth **media** on petri dishes. Only 1 millimeter long, its transparent nature permits an investigator to trace the origin and determine the exact anatomical position of any one of its 959 cells at any stage in development from fertilized egg to death. This has become known as “cell lineage.” Cell abnormalities, as well as abnormal migration of normal cells, are readily discovered, which has led to the identification and cataloging of a large number of mutants. The ability to directly see the individual sarcomeres (protein subunits) within a muscle cell quickly established *C. elegans* as a model system for the study of muscle structure and function. The gene for the muscle protein myosin, which could not be identified or cloned in other organisms, was readily cloned from *C. elegans*. This accomplishment, along with the study of the genes that give rise to “paralyzed” mutants of *C. elegans*, has had a major impact on the study of muscle development and diseases in humans. The transparency of *C. elegans* continues to expand the universality of this model system through the use of techniques that were not available in 1965, such as microinjection or laser microsurgery.

media nutrient source



somatic nonreproductive; not an egg or sperm

apoptosis programmed cell death

Insights from Laboratory Observation

Although anatomically and genetically simple, *C. elegans* mimics the life cycle of humans. Starting from a fertilized egg, it undergoes a complex development that gives rise to excretory, reproductive, digestive, and neuromuscular organ systems. The cell lineages of each of the 959 adult **somatic** cells have been directly observed, as has the fact that an additional unique set of 131 cells die during development. These deaths would have gone unnoticed, except for their demise was observable under the microscope. These observations led to the concept that **apoptosis** is a vital feature of development in multicellular organisms, and that it enables the shaping and carving of organs and tissues.

Today *C. elegans* remains one of the foremost model systems used in genetic research. Advanced descriptions have been made of its anatomy, cell lineages, developmental genetics, neural development, and reproductive cycle. The recent discovery of a mutation that doubles its life expectancy suggests that this “worm” will continue to expand its usefulness as a model system for the study of aging. Its growing significance as a member of the large, evolutionarily successful Nematoda phylum highlights the importance of *C. elegans* to the study of the genes involved in adaptive evolution. SEE ALSO APOPTOSIS; CLONING GENES; GENOME; MAPPING; MODEL ORGANISMS.

Diane C. Rein

Bibliography

Alberts, B., et al. *Molecular Biology of the Cell*, 3rd ed. New York: Garland Publishing, 1994.

Riddle, D. L., et al., eds. *C. elegans II*. Plainfield, NY: Cold Spring Harbor Laboratory Press, 1997.

Roberts, Leslie. “The Worm Project.” *Science* 248, no. 4962 (1990): 1310–1313.



Sanger, Fred

Molecular Biologist
1918–

Born August 13, 1918, in Rendcombe, Gloucestershire, United Kingdom, Fred Sanger has been breaking new ground in chemistry for decades. In fact, he is the only person to have won a Nobel Prize in chemistry twice, and is only one of four people ever to have won a Nobel Prize more than once.

While at Cambridge University in England he developed a new method for sequencing amino acids in proteins, which he used to identify the complete sequence of insulin. For this he was awarded his first Nobel Prize in chemistry in 1958. After this success, in 1961 Sanger moved to the MRC Laboratory of Molecular Biology, where he became head of the division of protein chemistry. His colleagues' interest in nucleic acids inspired him to turn his interest in sequencing to the research of nucleic acids.

In 1977 Sanger developed a sequencing method, called the “dideoxy” method, with which he determined the entire sequence of a bacterial virus called phi-X174. This was the first time a complete sequence of a DNA molecule had been established. For this achievement he was awarded the

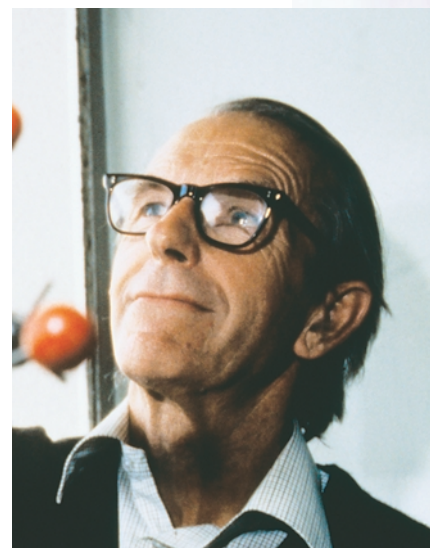
1980 Nobel Prize in chemistry, shared jointly with Walter Gilbert of Harvard University who had concurrently developed an alternative DNA sequencing method. Sanger's original sequence contained only 5,375 nucleotides, but his technology is now being used to determine sequences that are millions of nucleotides longer, including, importantly, the human genome. In his honor one of the major DNA sequencing centers in the world is named the Wellcome Trust Sanger Institute, at the Wellcome Trust Genome Campus, Cambridge, England. SEE ALSO SEQUENCING DNA.

Jeffery M. Vance

Bibliography

Internet Resource

"About Frederick Sanger." Wellcome Trust Sanger Institute. <<http://www.sanger.ac.uk/Info/Intro/sanger.shtml>>.



Fred Sanger shortly after receiving his second Nobel Prize for chemistry.

Science Writer

All good journalists are storytellers, and science writers are simply journalists who like to tell stories about science. Still, it has only been in the last few decades that science writing has become a profession in its own right, with journalists specifically trained to cover research ranging from genetics to particle physics.

The Growth of Specialization

Previously, most publications put so little emphasis on science that they did not need a specialized reporter. More and more, though, news organizations regard scientific results as necessary information, part of the everyday reporting of news. After all, science and technology have radically altered the way we live. For example, consider the development of antibiotics and vaccines, nuclear weapons, computer technologies, lasers, and fiber optics. Advances in science and technology will undoubtedly continue to occur in ways that we can not fully predict. The Human Genome Project, with all its promise and ethical unknowns, illustrates this perfectly.

The fast pace of scientific and technological advance has led to an increasing demand for science writers who understand science, and who can make others understand it as well. The membership of the National Association of Science Writers is now nearly 2,500. Science writing programs have sprung up at Boston University, Northwestern University, the University of California in Santa Cruz, the University of Maryland, and many other institutions. Most of these programs are aimed at journalists who wish to learn how to write about science, to more deftly translate jargon, explain complex experiments, and illuminate the people and the politics behind the science. A few, such as the program at Santa Cruz, are geared for science majors who wish to learn about journalism.

Breaking into the Field

While lack of science training is not a bar to working as a science journalist, a growing number of working science writers have undergraduate science degrees, or a combined degree in science and journalism. The bigger

publications and broadcast organizations place a premium on good writers with scientific training. Beyond that, being a trained researcher enables a journalist to ask questions that a colleague less knowledgeable might miss. Writers who do more specialized types of writing, such as medical writing for a physician audience, especially benefit from background in their field. Very few working science writers have advanced degrees, but there are advantages, both for the skills and knowledge gained and the credentials, which may lead to opportunities not otherwise available.

While valuing a scientific background, many media organizations put an even higher premium on good writing and the ability to make a difficult subject accessible to nonspecialist readers. A talented and hardworking journalist who is willing to do the necessary research and homework is thus often a valuable commodity even without specific science training.

Opportunities in Science Journalism

Opportunities for science writers continue to expand beyond such traditional media as newspapers, magazines, television, and radio. For example, there are on-line publications, some of which specialize only in research topics. Science writers are also hired as public information officers, to explain research at universities and government agencies. There are comparable jobs in private industry, working for specific companies or trade publications. Many of the big science associations publish their own journals or magazines and hire science-trained journalists for their news and comment sections.

The latter jobs mean writing for a science-literate audience. For the most part, however, science journalists explain a technical world to a non-technical audience. Their goal is not to teach people how to do the science, just how to appreciate it, evaluate it, and even enjoy it. Continually learning new science, and explaining it to those who are interested, is one of the great benefits of being a science writer.

Compensation

Beginning science writers may earn a starting salary of around \$20,000 to \$25,000, depending on their training, the type of job, the resources of the employer, and the region of the country. More experienced writers may earn between \$35,000 and \$60,000, and some writers earn even more. The highest salaries are paid to top-tier writers employed by major publications, or experienced medical writers working for pharmaceutical companies. SEE ALSO TECHNICAL WRITER.

Deborah Blum

Bibliography

- Blum, Deborah, and Mary Knudson, eds. *A Field Guide for Science Writers*. New York: Oxford University Press, 1998.
- Friedman, Sharon M., Sharon Dunwoody, and Carol L. Rogers, eds. *Communicating Uncertainty: Media Coverage of New and Controversial Science*. Mahwah, NJ: Lawrence Erlbaum Associates, 1999.
- Gastel, Barbara. *The Health Writers Handbook*. Ames: Iowa State University Press, 1998.

Internet Resources

American Association for the Advancement of Science: Media Relations. <<http://www.eurekalert.org>>.

National Association of Science Writers. <<http://www.nasw.org>>.

Society of Environmental Journalists. <<http://www.sej.org>>.

Selection

Selection is a process in which members of a population reproduce at different rates, due to either natural or human-influenced factors. The result of selection is that some characteristic is found in increasing numbers of organisms within the population as time goes on.

Types of Selection

Artificial selection, which is even older than agriculture, refers to a conscious effort to use for future breeding those varieties of a plant or animal that are most useful, attractive, or interesting to the breeder. Artificial selection is responsible for creating the enormous number of breeds of domestic dogs, for instance, as well as high-yielding varieties of corn and other agricultural crops.

Selection also occurs in nature, but it is not conscious. Charles Darwin called this natural selection. Darwin saw that organisms constantly vary in a population from generation to generation. He proposed that some variations allow an organism to be better adapted to a given environment than others in the population, allowing them to live and reproduce while others are forced out of reproduction by death, sterility, or isolation. These genetic variations gradually replace the ones that fail to survive or to reproduce. This gradual adjustment of the **genotype** to the environment is called adaptation. Natural selection was not only Darwin's key mechanism of evolution for the origin of species, it is also the key mechanism today for understanding the evolutionary biology of organisms from viruses to humans. Natural selection leads to evolution, which is the change in gene frequencies in a population over time.

The concept of selection plays an increasingly important role in biological theory. New fields such as evolutionary psychology rely heavily on natural selection to explain the evolution of human behavioral traits, such as mate choice, aggression, and other types of social behavior. A great difficulty in such a theoretically based science is the paucity of experimental or direct evidence for presumed past environments and presumed behavioral responses that were genetically adaptive.

Variation

The variation that selection requires arises from two distinct sources. The ultimate sources of variation are gene **mutation**, gene duplication and disruption, and chromosome rearrangements. Gene mutations are randomly occurring events that at a molecular level consist mostly of substitutions or small losses or gains of nucleotides within genes. Gene duplication makes new copies of existing genes, while gene disruptions destroy functional copies of genes, often through insertion of a mobile genetic element. Chromosome

genotype set of genes present

mutation change in DNA sequence





meiosis cell division that forms eggs or sperm

homozygotes individuals whose genetic information contains two identical copies of a particular gene

hybrid combination of two different types

rearrangements are much larger changes in chromosome structure, in which large pieces of chromosomes break off, join up, or invert. Individually, such mutations are rare. Most small mutations are either harmful or have no effect, and they may persist in a population for dozens or hundreds of generations before their advantages or disadvantages are evident.

The second source of variation arises from the shuffling processes undergone by genes and chromosomes during reproduction. During **meiosis**, maternally and paternally derived chromosome pairs are separated randomly, so that each sperm or egg contains a randomly chosen member of each of the twenty-three pairs. The number of possible combinations is over eight billion. Even more variation arises when pair members exchange segments before separating, in the process known as crossing over. The extraordinary variety in form exhibited even by two siblings is due primarily to the shuffling of existing genes, rather than to new mutations.

The Importance of the Environment

A disadvantageous trait in one environment may be advantageous in a very different environment. A classic example of this is sickle cell disease in regions where malaria is common. Individuals who inherit a copy of the sickle cell gene from both of their parents (**homozygotes**) die early from the disease, whereas heterozygotes (individuals who inherit only one copy of the gene) are favored in malarial areas (including equatorial Africa) over those without any copies, because they contract milder cases of malaria and thus are more likely to survive it.

Even though homozygotes rarely pass on their genes, because of their low likelihood of surviving to reproduce, the advantage of having one copy is high enough that natural selection continues to favor presence of the gene in these populations. Thus a malarial environment can keep the gene frequency high. However, in temperate regions where malaria is absent (such as North America), there is no heterozygote advantage to the sickle cell gene. Because heterozygotes still suffer from the disease, they are less likely to survive and reproduce. Thus, selection is gradually depleting the gene from the African American population that harbors it.

Artificial Selection

One of the first uses of genetic knowledge to improve yields and the quality of plant products was applied to **hybrid** seed production at the start of the twentieth century by George Shull. Artificial selection today is still done by hobbyists who garden or raise domestic animals. It is done on a more professional level in agriculture and animal breeding. The benefits are enormous. Virtually all commercial animal and plant breeding uses selection to isolate new combinations of traits to meet consumer needs. In these organisms, most of the variation is preexisting in the population or in related populations in the wild. The breeder's task is to combine (hybridize) the right organisms and select offspring with the desired traits.

In the antibiotic industry selection is used to identify new antibiotics. Usually, microorganisms are intentionally mutated to produce variation. Mutations can be induced with a variety of physical and chemical agents called

mutagens, which randomly alter genes. Some early strains of penicillin-producing molds were x-rayed and their mutations selected for higher yields.

Biologists also make use of selection in the process called molecular cloning. Here, a new gene is inserted into a host along with a marker gene. The marker is typically a gene for antibiotic resistance. To determine if the host has taken up the new genes, it is exposed to antibiotics. The ones who survive are those that took up the resistance gene, and so also have the gene of interest. This selection process allows the researcher to quickly isolate only those organisms with the new gene.

Selection in Humans

Both natural and artificial selection occur in human beings. If a trait is lethal and kills before reproductive maturity, then that gene mutation is gradually depleted from the population. Mutations with milder effects persist longer and are more common than very severe mutations, and **recessive** mutations persist for much longer than **dominant** ones. With a recessive trait, such as albinism, the parents are usually both carriers of a single copy of the gene and may not know that they carry it. If a child receives a copy of this gene from both of the carrier parents, the albino child may die young, may find it difficult to find a partner, or may end up marrying much later in life. This is usually considered a form of natural selection.

Considerable abuse of genetic knowledge in the first half of the twentieth century led to the **eugenics** movement. Advocates of eugenics claimed some people were more fit and others less fit (or unfit), and argued that the least fit should be persuaded or forced not to reproduce. Eugenicists typically defined as unfit those who were “feeble-minded, criminal, socially deviant, or otherwise undesirable.” Coerced sterilization, a form of artificial selection, was practiced on some of these individuals. SEE ALSO CLONING GENES; EUGENICS; HARDY-WEINBURG EQUILIBRIUM; MULLER, HERMANN; MUTAGENESIS; MUTATION.

Elof Carlson

Bibliography

Huxley, Julian. “Adaptation and Selection.” In *Evolution: The Modern Synthesis*. New York: Harper and Brothers, 1942.

Pianka, Eric. *Evolutionary Biology*, 6th ed. San Francisco: Addison-Wesley-Longman, 2000.

Sequencing DNA

A gene is a segment of DNA that carries the information needed by the cell to construct a protein. Which protein that is, when it is made, and how damage to it can give rise to genetic disease all depend on the gene’s sequence. In other words, they depend on how the building blocks of DNA, the **nucleotides** A, C, G, and T (adenine, cytosine, guanine, and thymine) are ordered along the DNA strand. For example, part of a gene may contain the base sequence TGGCAC, while part of another gene may contain the base sequence TCACGG. Knowing a gene’s base sequence can lead to

recessive requiring the presence of two alleles to control the phenotype

dominant controlling the phenotype when one allele is present

eugenics movement to “improve” the gene pool by selective breeding

nucleotides the building blocks of RNA or DNA

isolation of its protein product, show how individuals are related, or point the way to a cure for those people carrying it in its damaged form.

Overview

In 1977 two methods for sequencing DNA were introduced. One method, referred to as Maxam-Gilbert sequencing, after the two scientists at Harvard University who developed the technique, uses different chemicals to break radioactively labeled DNA at specific base positions. The other approach, developed by Frederick Sanger in England and called the chain termination method (also called the Sanger method), uses a DNA synthesis reaction with special forms of the four nucleotides that, when added to a DNA chain, stop (terminate) further chain growth.

By either method, a collection of single-stranded DNA fragments is produced, each fragment one base longer than the next. The length of a fragment depends on where a chemical cleaved the strand (in Maxam-Gilbert sequencing) or where a special terminator base was added (in the chain termination method). The fragments are then separated according to their size by a process called gel electrophoresis, in which the fragments are drawn through a gel material by electric current, with shorter fragments migrating through the gel faster than longer fragments. The DNA sequence is then “read” by noting which reaction produced which fragment.

Maxam-Gilbert Sequencing

All DNA sequencing protocols must incorporate a method for making the DNA fragments generated in the reaction “visible”; they must be capable of being detected. For Maxam-Gilbert sequencing a technique called end labeling is used, in which a radioactive atom is added to the ends of the DNA fragment being sequenced. The first step in this process is to use an **enzyme**, called a restriction endonuclease, to cut the DNA at a specific sequence. If the restriction endonuclease is Hind III, for example, the sequence AAGCTT will be cut.

The ends of the DNA segment made by the restriction endonuclease will have **phosphate groups** ($-\text{PO}_3^{2-}$) at its ends. An enzyme called a phosphatase is used to remove the phosphate group. Another enzyme, called a kinase, is then used to add a radioactive phosphate in its place. This reaction will add radioactive atoms onto both ends of the DNA restriction fragment. A second restriction endonuclease is used to make a cut within the end-labeled fragment and **gel electrophoresis** is then employed to separate the two resulting subfragments from each other. Each subfragment now has one labeled and one unlabeled end. The subfragment whose sequence is to be determined is cut out of the gel to purify it away from the other end-labeled subfragment.

The end-labeled piece of DNA is then divided and the fragments are placed in four separate tubes. They are then treated with different chemicals that weaken and break the bond holding the base to the backbone of the DNA molecule. These chemicals are base specific. In other words, one chemical causes the “C” reaction, in which the bond holding the C base in position is broken. Another chemical breaks the bond holding the G in place (the “G” reaction). Another breaks both G and A bases from the DNA back-

enzyme a protein that controls a reaction in a cell

phosphate groups PO_4 groups, whose presence or absence often regulates protein action

gel electrophoresis technique for separation of molecules based on size and charge

bone (the “G+A” reaction), and a fourth breaks the bonds holding the C and T bases in place (the “C+T” reaction).

Each reaction is limited so that each DNA molecule will have only one of its base positions altered. Within the “C” reaction, for example, each DNA molecule will have only one of its C bases weakened and removed. One of the DNA molecules may have the C base closest to the end dislodged. On another DNA molecule, a C that appears 500 bases away from the end may be removed. Every C position in the population of molecules, however, is subject to treatment.

When the first step in the reaction is concluded, another reagent is then used to completely break the DNA strands at the points where the bases have been removed. In this way, a collection of DNA fragments is generated that differ in size according to the position along the strand where the break occurred.

Each reaction is electrophoresed in its own lane on a gel that separates the DNA fragments by their length. A process called autoradiography is then used to detect the separated fragments. In this technique, X-ray sensitive film is placed flat against the gel under conditions of complete darkness. Since the fragments made in the sequencing reactions are end-labeled with radioactive atoms, their emissions will expose the X-ray film at the positions where they are found on the gel. When the X-ray film is developed, bands, like a bar code, reveal the sizes of the fragments generated in each separate reaction.

Each band that has been rendered visible on the X-ray film differs from the one above or the one below it by a single base. The DNA sequence is then read from the bottom of the gel upward. A band found in the “G” reaction lane is read as a G, a band found in the “C” lane is read as a C, a band found in the lane from the “G+A” reaction but with no corresponding band of the same length in the “G” lane is read as an A, and a band in the “C+T” reaction without a corresponding base in the “C” lane is read as a T (see Figure 1).

Chain Termination Method

Because it employs fewer steps and does not require the use of **restriction enzymes**, the chain termination method of DNA sequencing is used in more laboratories than is the Maxam-Gilbert approach. Chain termination sequencing is a clever variation of the reaction used to replicate DNA, and requires only a handful of components. Four reaction tubes, designated “A,” “C,” “G,” and “T,” are prepared and the DNA strand whose sequence is to be determined is added into each tube. This DNA strand is called the template.

Along with the template, a short, single-stranded piece of DNA (called a primer) is added; it attaches specifically to one section of the template and serves as a starting point for synthesis of a new DNA strand. Also added are the four “building block” nucleotides, a **buffer** (to maintain the proper pH level), the enzyme DNA polymerase, and its cofactor magnesium, all of which are needed to extend the primer into a full-length DNA chain.

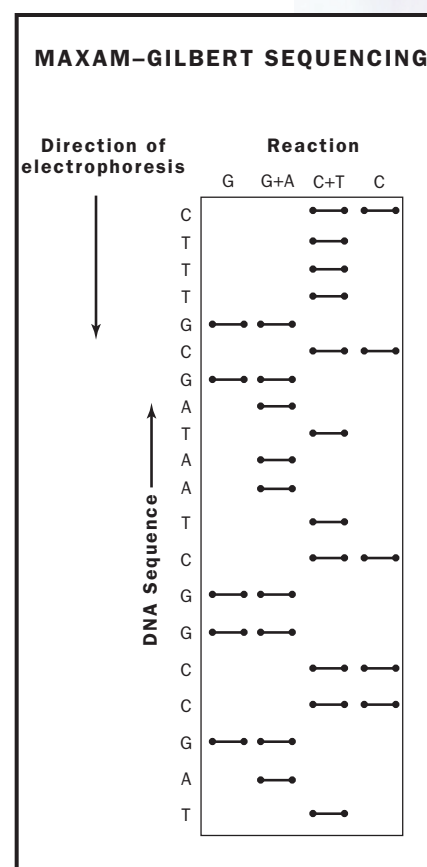


Figure 1. Schematic diagram of autoradiogram showing bands generated by the Maxam-Gilbert method of sequencing DNA. The DNA sequence is read from the bottom of the gel by noting which gel lane contains a band.

restriction enzymes
enzymes that cut DNA at a particular sequence

buffers substances that counteract rapid or wide pH changes in a solution